SQL Business Analysis Report: Online Bookstore Sales Database

Project Overview

This project presents a comprehensive business analysis of an Online Bookstore database using SQL. The objective is to transform raw transactional data into meaningful business insights that support data-driven decision-making.

The analysis focuses on three major business areas:

- **Sales Performance**
- **Customer Analysis**
- **Inventory Management**

By applying SQL queries to relational data, this project uncovers sales trends, customer purchasing behavior, revenue distribution, and inventory opportunities. The project simulates real-world business scenarios commonly encountered by Data Analysts and demonstrates how SQL can be used to answer strategic business questions.

Dataset Description

The database consists of three relational tables that together represent the bookstore's operations.

Table Name	Description	Columns Included
Books	Contains information about every book available in the bookstore.	Book ID, Title, Author, Genre, Price, Stock, Publication Details
Customers	Contains customer information.	Customer ID, Customer Name, City, Country, Contact Information

Table Name	Description	Columns Included
Orders	Contains all transactional sales records.	Order ID, Customer ID, Book ID, Order Date, Quantity Purchased, Total Amount

The relationships between these tables allow business information to be extracted through SQL joins, aggregations, filtering, and advanced querying techniques.

Project Objectives

The primary objectives of this project are to:

- Analyze bookstore sales performance using SQL.
- Understand customer purchasing behavior.
- Evaluate inventory health and identify restocking opportunities.
- Demonstrate practical SQL skills on a relational database.
- Apply industry-standard SQL concepts to solve business problems.
- Generate actionable insights that support business decision-making.
- Build a portfolio-ready SQL project that showcases both technical and analytical capabilities.

Business Analysis Report

1. Sales Performance Analysis

The sales analysis focused on understanding which books, genres, and authors contribute most to the bookstore's revenue.

Key Findings:

- The bookstore's best-selling titles were identified by analyzing the total quantity sold for each book. These books represent the products with the strongest customer demand and are the primary contributors to overall sales volume.
- Genre-wise revenue analysis revealed that Romance is the highest revenue-generating genre within the bookstore. This indicates a strong customer preference for Romance titles and highlights the genre as one of the company's most profitable product categories.
- Revenue generated by individual authors was also analyzed, allowing the business to

identify authors who contribute the most to overall sales. Such insights can support future purchasing decisions, promotional campaigns, and author partnerships.
- Book-level revenue analysis showed that a relatively small number of titles account for a significant share of total revenue. These high-performing books represent valuable assets that should remain consistently available to customers.
- The analysis also identified several books that generated above-average revenue despite selling fewer copies than average. This suggests the presence of higher-priced or premium books capable of generating substantial revenue without requiring large sales volumes.

Business Impact:

These insights enable the business to:

- Identify top-performing books.
- Focus marketing efforts on profitable genres.
- Strengthen relationships with high-performing authors.
- Optimize pricing strategies.
- Increase revenue through targeted promotions.

2. Customer Analysis

Customer analysis was conducted to better understand purchasing behavior, customer value, and geographical sales distribution.

Key Findings:

- The analysis identified Kim Turner as the highest-spending customer based on total purchase value. High-value customers such as this represent important long-term assets and may benefit from loyalty programs or personalized marketing campaigns.
- Revenue analysis by country showed that Cambodia generated the highest overall revenue among all countries represented in the dataset. This indicates a particularly valuable regional market that may deserve additional marketing investment.
- Order frequency analysis revealed that while most customers place relatively few orders, a smaller group consistently places significantly more orders than average. These repeat customers contribute substantially to overall business performance and represent highly engaged buyers.
- Customer distribution across cities highlighted locations with larger customer populations, providing useful information for regional marketing initiatives and future expansion opportunities.
- Finally, customers purchasing more books than the average customer were identified,

helping distinguish the bookstore's most loyal and valuable customer segment.

Business Impact:

These findings can help the business:

- Identify and retain high-value customers.
- Develop customer loyalty programs.
- Improve regional marketing strategies.
- Better understand purchasing behavior.
- Increase customer lifetime value.

3. Inventory & Business Insights

Inventory analysis focused on stock availability, product performance, and opportunities for inventory optimization.

Key Findings:

- Several books were found to have very low or zero stock levels, indicating products that may require immediate replenishment to avoid stockouts and lost sales.
- Revenue generated by individual authors highlighted those making the greatest financial contribution to the bookstore, providing additional support for future inventory planning and purchasing decisions.
- The analysis also identified books that have never been ordered. These products may require promotional campaigns, pricing adjustments, or inventory review to improve turnover.
- Inventory distribution by genre showed that Science Fiction contains the largest number of books currently held in inventory, indicating that this category receives significant stocking priority.
- Finally, books exhibiting high sales but relatively low inventory levels were identified as strong candidates for immediate restocking. These products represent the highest risk of lost revenue if inventory shortages occur.

Business Impact:

Inventory insights help the business:

- Prevent stock shortages.
- Improve inventory planning.
- Reduce slow-moving inventory.
- Optimize warehouse utilization.
- Prioritize restocking decisions.

- Improve product availability.

SQL Concepts Demonstrated

Throughout this project, a wide range of SQL concepts were applied to solve real-world business problems.

Category	SQL Techniques & Concepts Applied
Data Retrieval & Filtering	SELECT, DISTINCT, Column Aliasing, WHERE, Comparison Operators, Logical Operators (AND, OR), ORDER BY, LIMIT
Aggregation & Grouping	SUM(), COUNT(), AVG(), ROUND(), GROUP BY
Table Joins & Advanced SQL	INNER JOIN, LEFT JOIN, Subqueries, Common Table Expressions (CTEs)
Business Analysis Techniques	Sales & Revenue Analysis, Customer Behavior Analysis, Inventory Restocking Identification, Business-Oriented Data Exploration

Overall Business Insights

The SQL analysis demonstrates how transactional data can be transformed into meaningful business intelligence.

The bookstore's revenue is primarily driven by a small group of high-performing books and the Romance genre, while customer analysis highlights the importance of retaining high-value and repeat customers. Inventory evaluation reveals opportunities to improve stock management by restocking fast-selling books and reviewing products that have never generated sales.

Collectively, these findings provide valuable guidance for improving sales performance, customer engagement, inventory planning, and overall business efficiency.

Conclusion

This project demonstrates the practical application of SQL in solving real-world business problems using a relational database. By analyzing sales performance, customer behavior, and inventory management, the project illustrates how SQL can transform raw data into actionable business insights.

Across 15 business scenarios, essential SQL concepts—including Joins, Aggregate Functions, GROUP BY, Subqueries, Common Table Expressions (CTEs), Filtering, Sorting, and Data Aggregation—were applied to answer meaningful business questions.

Beyond demonstrating technical SQL proficiency, the project emphasizes analytical thinking by interpreting query results and translating them into business recommendations. The insights generated can support decision-making related to product strategy, customer retention, revenue optimization, and inventory management.

Overall, this project showcases both technical and business-oriented data analysis skills, making it a strong portfolio piece that reflects the practical responsibilities of a Data Analyst working with relational databases.